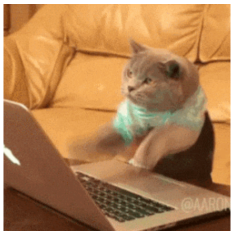
Last updated in Apr 2026



Luca Domeniconi

AI Researcher graduated in Artificial Intelligence at the University of Bologna

luca24ever@gmail.com | liuktc.github.io | LinkedIn | liuktc

Education

M.Sc. **University of Bologna**, Artificial Intelligence — Bologna, Italy, Sept 2023 – Mar 2026
- Thesis: *Using Sparse Features to Characterize and Mitigate Concept-Dependent Modality Gap in Vision-Language Models*
- Grade: 110/110 *with honors*

B.Sc. **University of Bologna**, Computer Engineering — Bologna, Italy, Sept 2020 – Oct 2023
- Grade: 109/110

Publications

How to Evaluate and Refine your CAM — Apr 2026
Luca Domeniconi, Alessandra Stramiglio, Michele Lombardi, Samuele Salti
10.1234/neurips.2023.1234 (ICPR 2026)

Experience

EPFL, Research Intern (Master's Thesis) — Lausanne, Switzerland, Oct 2025 – Mar 2026, 6 months
- Conducted research on interpretability and vision-language models for my Master's thesis project in IDIAP Lab, under the supervision of Prof. Andrea Cavallaro

Belluzzi Fioravanti High School, Computer Science Teacher — Bologna, Italy, Jan 2024 – Feb 2024, 2 months
- Taught students about computer architecture and networking using sockets

Gruppo Finmatica, Software Engineer Intern — Bologna, Italy, Mar 2023 – June 2023, 4 months
- Deployment of a web based collaborative document editor and development of a React Native prototype for barcode scanning to assist warehouse personnel in inventory management

Private Tutor — Rimini, Italy, 2019 – 2025, 6 years
- Private computer science and mathematics lessons for undergratuate and high school students

Achievements

1st Place at EPFL LauzHack 2024, BMS Challenge — Lausanne, Switzerland
Developed an explainable time-series forecasting model using Gaussian Processes and Shapley values to predict pharmaceutical demand, ranking 1st out of 12 teams in the Bristol Myers Squibb 24-hour hackathon challenge.

1st Place at Tablut AI Agent Challenge — University of Bologna
Designed and developed an AI player for the asymmetric board game Tablut using adversarial search and heuristic evaluation, ranking 1st out of 10 agents.

Selected Projects

ICD Multimodal Diagnosis — *Project Repository*
Implemented a Tree-constrained multimodal ICD-9 generation pipelinefrom chest X-rays and clinical text using custom made VLMs

Semantic Segmentation of Unexpected Objects on Roads — *Project Repository*
Implemented an open-world semantic segmentation model using a pre-trained DINOv3 plus metric learning to segment known classes while detecting anomalous regions.

Constraint Programming for Multiple Couriers Optimization — *Project Repository*
Modelled the Capacitated Vehicle Routing Problem using CP, SAT, SMT, and MILP (MiniZinc, OR-Tools, Z3, Gurobi)

Shelves Product Detection *Project Repository*
Instance detection of products on shelves using only classical computer vision techniques with OpenCV

Skills

Research Interests: Explainability, Attribution Mehthods, Vision-Language Models
Computer Vision: Python, OpenCV, PyTorch
Machine Learning: Keras, numpy, pandas, scikit-learn, Matplotlib
Web and Mobile Development: JavaScript, NodeJS, React, React Native, Android Native
Programming Languages and Tools: Java, C, C#, \LaTeX, SQL, MATLAB, Minizinc, Unity 3D, Docker, Blender